

LECTURE 6

Monotone Convergence, the Squeeze Theorem, and the Upper and Lower Limits

MATH S4061 | Introduction to Modern Analysis I

Thursday, 11 June 2026

Of the small there is no smallest, but always a smaller; and of the large there is always a larger.

ANAXAGORAS, FRAGMENT B3 (5TH C. BCE)

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Overview. Picking up from completeness, this lecture assembles the basic machinery for limits of real sequences. Monotonicity plus boundedness already forces convergence; the comparison and squeeze principles let limits be read off from neighbours; and for sequences that oscillate, the upper and lower limits $\limsup$ and $\liminf$ record the band the sequence is eventually trapped in, coinciding exactly when the sequence converges. After recording four limits that recur throughout the subject, we open the theory of series: convergence through partial sums, the Cauchy criterion, and the first divergence test. The companion text is Rudin, Chapter 3, especially $\text{R } 3.12\text{--}3.40$.

6.1 Review of Completeness and Cauchy Sequences

Definition 6.1.1

Complete metric space

A metric space (X, d) is *complete* (R 3.12) if every Cauchy sequence in X converges to a point of X .

Example 6.1.2

Complete and incomplete spaces

- Compact metric spaces are complete (R 3.11).
- $\mathbb{R}^k$ is complete (R 3.11).
- $\mathbb{Q}$ and $(0, 1)$ are *not* complete (cf. R 3.12).

Remark 6.1.3

Testing convergence without the limit

In a complete space — in particular in $\mathbb{R}^k$ — it is enough to check whether a sequence is Cauchy; there is no need to know the limit in advance. In an incomplete space such as $\mathbb{Q}$ this fails: a Cauchy sequence may have no limit (6.1.2; cf. R 3.11–R 3.12).

6.2 Monotone Sequences

Definition 6.2.1

Monotonic sequences

A sequence $\{s_n\}$ is monotonic in Rudin’s sense (R 3.13) as follows:

- *monotonic increasing* if $s_n \leq s_{n+1}$ for all n ;
- *monotonic decreasing* if $s_n \geq s_{n+1}$ for all n .

“Monotonic” means either increasing or decreasing.

Theorem 6.2.2

Monotone convergence

If $\{s_n\} \subseteq \mathbb{R}$ is monotonic and bounded, then $\{s_n\}$ converges (R 3.14).¹

Proof 6.2.2

Proof by a Supremum Argument of the Monotone Convergence Theorem.

- (1) Treat the increasing case (the decreasing case is analogous). Since $\{s_n\}$ is bounded above, the least-upper-bound property of $\mathbb{R}$ provides

$$s := \sup_n s_n.$$

¹This is the first place completeness earns its keep by *manufacturing* a limit rather than asking you to guess one. A monotone sequence can head only one way, and boundedness blocks the exit, so the terms have nowhere to go but to pile up against their least upper bound—and that bound, $\sup_n s_n$, *is* the limit. Read the two hypotheses as a division of labour: monotone forbids oscillation, bounded forbids escape to infinity.

- (2) Let $\varepsilon > 0$. Because $s - \varepsilon < s$, the number $s - \varepsilon$ is not an upper bound of $\{s_n\}$, so there is N_ε with

$$s_{N_\varepsilon} > s - \varepsilon.$$

- (3) As the sequence is increasing and s is an upper bound, for every $n \geq N_\varepsilon$

$$s - \varepsilon < s_{N_\varepsilon} \leq s_n \leq s,$$

hence $|s_n - s| < \varepsilon$. Therefore $s_n \rightarrow s$. ■

[A5] cf. R 3.14; the decreasing case follows by applying the increasing case to $\{-s_n\}$.

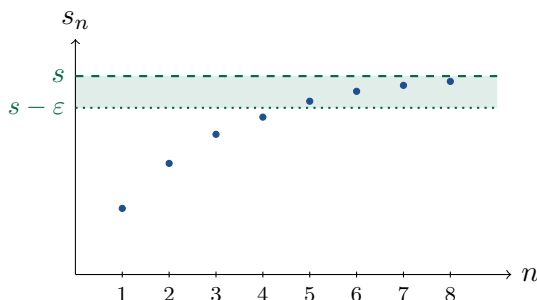


Figure 6.2.3. CLIMBING TO THE SUPREMUM. An increasing sequence bounded above closes in on $s = \sup_n s_n$: the terms rise but never cross the dashed bound, and from some N_ε on they all lie in the band $(s - \varepsilon, s]$.

6.3 Comparison and Squeeze

Proposition 6.3.1

Order is preserved in the limit

Let $\{s_n\}$ and $\{t_n\}$ be convergent real sequences with $s_n \leq t_n$ for all n . Then

$$\lim_{n \rightarrow \infty} s_n \leq \lim_{n \rightarrow \infty} t_n.$$

This is the ordinary-limit form of Rudin's comparison theorem for upper and lower limits (cf. R 3.19).

Idea. Suppose instead that $s := \lim s_n$ and $t := \lim t_n$ satisfy $s > t$. Separate s and t by disjoint ε -balls, with $\varepsilon < \frac{1}{2}|s - t|$, so that $B_\varepsilon(t)$ and $B_\varepsilon(s)$ sit on opposite sides of the midpoint $\frac{s+t}{2}$.

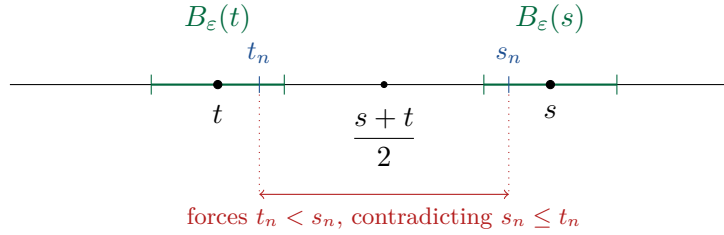


Figure 6.3.2. SEPARATING THE LIMITS. With $\varepsilon < \frac{1}{2}|s - t|$, the balls $B_\varepsilon(t)$ and $B_\varepsilon(s)$ fall on opposite sides of the midpoint $\frac{s+t}{2}$, which is impossible once $s_n \leq t_n$.

Proof 6.3.1

Proof by Contradiction that Order Passes to the Limit.

- (1) Write $s = \lim s_n$, $t = \lim t_n$, and suppose for contradiction $s > t$. Since $s - t > 0$, choose

$$0 < \varepsilon < \frac{1}{2}|s - t| = \frac{1}{2}(s - t).$$

- (2) By convergence there are N_1, N_2 with $|s_n - s| < \varepsilon$ for $n \geq N_1$ and $|t_n - t| < \varepsilon$ for $n \geq N_2$. With $N = \max\{N_1, N_2\}$, for every $n \geq N$

$$s_n > s - \varepsilon \quad \text{and} \quad t_n < t + \varepsilon.$$

- (3) The choice of ε puts these on opposite sides of the midpoint:

$$t + \varepsilon < t + \frac{1}{2}(s - t) = \frac{s + t}{2} = s - \frac{1}{2}(s - t) < s - \varepsilon.$$

- (4) Chaining the inequalities, for every $n \geq N$,

$$t_n < t + \varepsilon < \frac{s + t}{2} < s - \varepsilon < s_n,$$

so $t_n < s_n$, contradicting $s_n \leq t_n$. Hence $s \leq t$, i.e. $\lim s_n \leq \lim t_n$. ■

[A9]

Proposition 6.3.3

Squeeze theorem

Suppose $r_n \leq s_n \leq t_n$ for all n , where $r_n \rightarrow \ell$ and $t_n \rightarrow \ell$. Then $s_n \rightarrow \ell$. This is the comparison principle used in Rudin's special-sequence computations (cf. R 3.20).

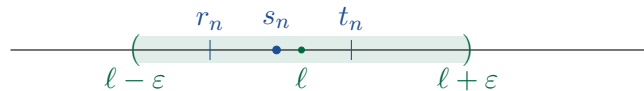


Figure 6.3.4. THE SQUEEZE. Once r_n and t_n lie in $(\ell - \varepsilon, \ell + \varepsilon)$, the trapped term s_n must lie there too.

Proof 6.3.3

Proof by an ε - N Estimate of the Squeeze Theorem.

(1) Let $\varepsilon > 0$. Since $r_n \rightarrow \ell$ and $t_n \rightarrow \ell$, there is N with, for all $n \geq N$,

$$|r_n - \ell| < \varepsilon \quad \text{and} \quad |t_n - \ell| < \varepsilon,$$

so $r_n, t_n \in (\ell - \varepsilon, \ell + \varepsilon)$.

(2) Using $r_n \leq s_n \leq t_n$, for every $n \geq N$

$$\ell - \varepsilon < r_n \leq s_n \leq t_n < \ell + \varepsilon,$$

hence $|s_n - \ell| < \varepsilon$. As $\varepsilon > 0$ was arbitrary, $s_n \rightarrow \ell$.

[A2]

■

6.4 The Upper and Lower Limits

Convergence $s_n \rightarrow \ell$ says that for every $\varepsilon > 0$ there is N with $s_n \in (\ell - \varepsilon, \ell + \varepsilon)$ for $n \geq N$. For a bounded sequence that does *not* converge, no single ℓ works. Instead there are two values $\ell^- \leq \ell^+$ such that the sequence is eventually trapped in any ε -enlargement of the band $[\ell^-, \ell^+]$:

$$\forall \varepsilon > 0 \quad \exists N \text{ s.t. } s_n \in (\ell^- - \varepsilon, \ell^+ + \varepsilon) \quad \text{for } n \geq N.$$

We write $\ell^- = \liminf s_n$ and $\ell^+ = \limsup s_n$.

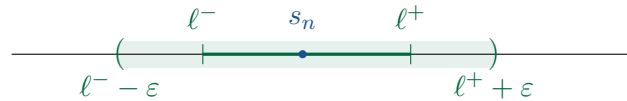


Figure 6.4.1. THE TRAPPING BAND. For a bounded sequence, every ε -enlargement of $[\ell^-, \ell^+]$ eventually contains all the terms.

The two coincide exactly when the sequence converges:

$$s_n \rightarrow \ell \iff \liminf s_n = \limsup s_n = \ell.$$

Example 6.4.2

An oscillating sequence

For $s_n = (-1)^n \frac{n}{n-1}$ (with $n \geq 2$), the even terms approach $+1$ from above and the odd terms approach -1 from below. Hence $\liminf s_n = -1$ and $\limsup s_n = +1$, and the sequence does not converge, since these differ (cf. R 3.18).

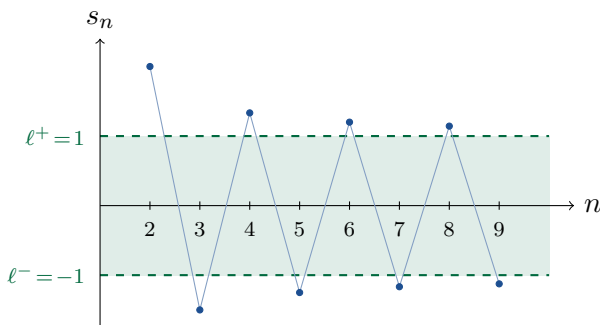


Figure 6.4.3. TWO LIMIT POINTS. The oscillating sequence $s_n = (-1)^n \frac{n}{n-1}$ has no limit: the even terms fall toward $\ell^+ = \limsup s_n = 1$ from above and the odd terms rise toward $\ell^- = \liminf s_n = -1$ from below. Every ε -enlargement of the shaded band $[\ell^-, \ell^+]$ eventually traps the whole tail.

Definition 6.4.4

Divergence to $\pm\infty$

For a real sequence $\{t_n\}$ (R 3.15),

- $t_n \rightarrow +\infty$ means: for every $M > 0$ there is N with $t_n > M$ for all $n \geq N$;
- $t_n \rightarrow -\infty$ means: for every $M > 0$ there is N with $t_n < -M$ for all $n \geq N$.

Such sequences diverge, but in a specific, controlled way.

Definition 6.4.5

Limit inferior and limit superior

The *upper* and *lower limits* of a real sequence $\{s_n\}$ are defined by the tail extrema

$$\limsup s_n = \lim_{n \rightarrow \infty} \left(\sup_{k \geq n} s_k \right), \quad \liminf s_n = \lim_{n \rightarrow \infty} \left(\inf_{k \geq n} s_k \right).$$

The tails are nested, $\{s_k : k \geq n+1\} \subseteq \{s_k : k \geq n\}$, so $\sup_{k \geq n} s_k$ is non-increasing in n (and $\inf_{k \geq n} s_k$ non-decreasing); hence both limits always exist in the extended reals $[-\infty, +\infty]$.² Two characterizations follow.

(1) Subsequential limits. If $\{s_n\}$ is bounded and E is its set of subsequential limits, then

$$\liminf s_n = \min E, \quad \limsup s_n = \max E.$$

²Picture it this way: $\sup_{k \geq n} s_k$ is the ceiling over the tail from n onward, and discarding more early terms can only lower a supremum, so these ceilings march downward to a limit—the lim sup. It is the highest value the sequence keeps crowding back up to, while lim inf is the matching floor. A convergent sequence has ceiling and floor equal; a bounded oscillating one holds them apart, pinning the tail between two honest numbers (6.4.2).

(2) Eventual trapping (finite $\liminf$, $\limsup$). For every $\varepsilon > 0$ there is N_ε with

$$s_n \in (\liminf s_n - \varepsilon, \limsup s_n + \varepsilon) \quad \text{for } n \geq N_\varepsilon,$$

and $[\liminf s_n, \limsup s_n]$ is the smallest closed interval with this property.

This tail-extrema presentation is equivalent to Rudin's subsequential-limit definition (R 3.16) and characterization (R 3.17).

Remark 6.4.6

The extreme limits are attained

For a *bounded* sequence, E is closed, bounded, and nonempty, so it contains its own extrema: $\inf E \in E$ and $\sup E \in E$. Hence $\liminf s_n$ and $\limsup s_n$ are themselves subsequential limits of $\{s_n\}$. (For an unbounded sequence this holds in the extended reals, with $\pm\infty$ admitted as subsequential limits.) Compare R 3.17.

Remark 6.4.7

Existence

If $\{s_n\}$ is bounded, then E is bounded and nonempty, so $\liminf s_n$ and $\limsup s_n$ exist as finite reals. If $\{s_n\}$ is unbounded, then E may include $+\infty$ and/or $-\infty$, and $\liminf s_n, \limsup s_n$ may equal $\pm\infty$ (cf. R 3.16–R 3.17).

6.5 Some Standard Limits

Proposition 6.5.1

Four standard limits

- (a) If $|x| < 1$, then $x^n \rightarrow 0$.
- (b) If $p > 0$, then $\frac{1}{n^p} \rightarrow 0$.
- (c) If $p > 1$, then $\sqrt[p]{p} \rightarrow 1$.
- (d) $\sqrt[n]{n} \rightarrow 1$.

These are the recurring special limits collected in R 3.20(a)–(c),(e).

Proof 6.5.1.a

Proof by the Binomial Inequality of $x^n \rightarrow 0$.

- (1) If $x = 0$ then $x^n = 0 \rightarrow 0$, so assume $0 < |x| < 1$. Then $1/|x| > 1$, so $|x| = \frac{1}{1+p}$ for a unique $p > 0$ (the map $p \mapsto \frac{1}{1+p}$ is a bijection $(0, \infty) \rightarrow (0, 1)$).
- (2) By the binomial theorem, keeping the nonnegative $k = 0, 1$ terms,

$$(1+p)^n = \sum_{k=0}^n \binom{n}{k} p^k \geq 1 + np.$$

(3) Hence

$$|x|^n = \frac{1}{(1+p)^n} \leq \frac{1}{1+np} \xrightarrow{n \rightarrow \infty} 0,$$

and since $|x^n| = |x|^n \rightarrow 0$, we conclude $x^n \rightarrow 0$. ■

[A2]

Proof 6.5.1.b

Proof by an ε - N Estimate of $1/n^p \rightarrow 0$.

(1) Let $\varepsilon > 0$ and choose N with $N > (1/\varepsilon)^{1/p}$, equivalently $\frac{1}{N^p} < \varepsilon$.

(2) Then for every $n \geq N$,

$$\frac{1}{n^p} \leq \frac{1}{N^p} < \varepsilon,$$

so $\frac{1}{n^p} \rightarrow 0$. ■

[A2]

Proof 6.5.1.c

Proof by the Squeeze Theorem of $\sqrt[p]{p} \rightarrow 1$.

(1) Assume $p > 1$ and set $x_n = \sqrt[p]{p} - 1 \geq 0$. By the binomial inequality from (a),

$$1 + nx_n \leq (1 + x_n)^n = p,$$

so $0 \leq x_n \leq \frac{p-1}{n}$.

(2) Since $\frac{p-1}{n} \rightarrow 0$, the squeeze theorem (6.3.3) gives $x_n \rightarrow 0$, i.e. $\sqrt[p]{p} = 1 + x_n \rightarrow 1$. ■

[A2] uses 6.3.3.

Proof 6.5.1.d

Proof by the Squeeze Theorem of $\sqrt[n]{n} \rightarrow 1$.

(1) Set $x_n = \sqrt[n]{n} - 1 \geq 0$, so $n = (1 + x_n)^n$. The linear term is too weak, so keep the $k = 2$ term:

$$n = (1 + x_n)^n \geq \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} x_n^2 \quad (n \geq 2).$$

(2) Rearranging, $x_n^2 \leq \frac{2}{n-1}$, hence $0 \leq x_n \leq \sqrt{\frac{2}{n-1}}$.

(3) Since $\sqrt{\frac{2}{n-1}} \rightarrow 0$, the squeeze theorem (6.3.3) gives $x_n \rightarrow 0$, i.e. $\sqrt[n]{n} = 1 + x_n \rightarrow 1$. ■

[A2] keep the $k = 2$ binomial term; uses 6.3.3.

Remark 6.5.2

The case $0 < p < 1$ in (c)

For $0 < p < 1$, apply part (c) to $\sigma = 1/p > 1$: since $\sqrt[p]{\sigma} \rightarrow 1$, also $\sqrt[p]{p} = 1/\sqrt[p]{\sigma} \rightarrow 1$ (and $\sqrt[p]{1} = 1$).

Hence $\sqrt[p]{p} \rightarrow 1$ for every $p > 0$; the full $p > 0$ statement is R 3.20(b).

6.6 Introduction to Series

(Not covered on the midterm.)

As a field, $\mathbb{R}$ hands us directly only the polynomials $1, x, x^2, \dots$ and, as their quotients, the rational functions $P(x)/Q(x)$. What about e^x , $\sin x$, $\cos x$? These are reached not by field operations but by infinite sums — for instance

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}.$$

To make sense of such a sum we must first say what an infinite sum *is*.

Definition 6.6.1

Series and partial sums

Given a sequence $\{a_n\}$ in $\mathbb{R}$, its k -th *partial sum* is

$$s_k = a_1 + \cdots + a_k = \sum_{i=1}^k a_i \quad (s_0 = 0, \text{ the empty sum}).$$

If $s_k \rightarrow s$, the *series* $\sum a_n$ *converges*, and we write

$$\sum_{i=1}^{\infty} a_i = \lim_{k \rightarrow \infty} s_k = s.$$

A series may instead be indexed from any starting point—often $n = 0$, as for $\sum c_n x^n$ —with the partial sums adjusted accordingly.³ This is Rudin’s series convention (R 3.21) with the real-valued specialization used in lecture.

Theorem 6.6.2

Cauchy criterion for series

A series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is N such that

$$\left| \sum_{k=m}^n a_k \right| < \varepsilon \quad \text{for all } n \geq m \geq N.$$

³A series introduces no genuinely new operation; there is no act of adding infinitely many numbers at once. The symbol $\sum a_n$ is shorthand for a *sequence*—the partial sums s_k —together with its limit, so every question about a series is a question about that sequence, and every tool for sequences (monotonicity, the Cauchy criterion, the algebra of limits) carries straight over.

This is the Cauchy criterion for series (R 3.22).

Proof 6.6.2

Proof by Completeness of $\mathbb{R}$ of the Cauchy Criterion.

- (1) For $n \geq m \geq 1$ one has $s_n - s_{m-1} = \sum_{k=m}^n a_k$ (with $s_0 = 0$), so the displayed condition says precisely that the partial sums $\{s_k\}$ form a Cauchy sequence.
- (2) Since $\mathbb{R}$ is complete (6.1.1), $\{s_k\}$ converges if and only if it is Cauchy. Hence $\sum a_n$ converges if and only if the criterion holds.

■

[Direct] cf. R 3.22; the partial sums converge iff they are Cauchy, and $\mathbb{R}$ is complete.

Corollary 6.6.3

Vanishing of the terms

If $\sum a_n$ converges, then $a_n \rightarrow 0$ (R 3.23).⁴

Proof 6.6.3

Proof by the Cauchy Criterion with $m = n$.

- (1) Take $m = n$ in the criterion (6.6.2): for every $\varepsilon > 0$ there is N with

$$|a_n| = \left| \sum_{k=n}^n a_k \right| < \varepsilon \quad \text{for all } n \geq N.$$

- (2) Therefore $a_n \rightarrow 0$.

■

[Direct] cf. R 3.23.

So $a_n \rightarrow 0$ is *necessary* for convergence. Is it *sufficient*? No.

Proposition 6.6.4

The harmonic series diverges

Although $\frac{1}{n} \rightarrow 0$, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. This is the $p = 1$ case of Rudin's p -series theorem (R 3.28).

Proof 6.6.4

Proof by Dyadic Grouping of the Divergence of the Harmonic Series.

- (1) Round each term down to the next power of $\frac{1}{2}$: replace $\frac{1}{3}$ by $\frac{1}{4}$; then $\frac{1}{5}, \frac{1}{6}, \frac{1}{7}$ by $\frac{1}{8}$; and so on. If s'_k denotes the resulting partial sums, then $s_k \geq s'_k$ for every k .

⁴Use this in one direction only. It says convergence *forces* the terms to die out, so if $a_n \not\rightarrow 0$ the series cannot converge—a quick divergence test. It does not say the reverse: terms dying out is not enough, as the harmonic series shows next (6.6.4). The terms must not merely vanish but vanish fast enough to be summable.

- (2) The rounded series groups into dyadic blocks of equal weight:

$$1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots,$$

each block summing to $\frac{1}{2}$; hence $s'_{2^m} = 1 + \frac{m}{2}$.

- (3) Therefore $s_{2^m} \geq s'_{2^m} = 1 + \frac{m}{2} \rightarrow \infty$, so the partial sums $\{s_n\}$ have a subsequence diverging to $+\infty$ and cannot converge. Thus $\sum \frac{1}{n}$ diverges. ■

[Direct] Cauchy condensation; cf. R 3.27–R 3.28.

Proposition 6.6.5

The p -series

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$. Rudin proves the full convergence/divergence dichotomy in R 3.28.

Proof 6.6.5

Proof of the Convergence of the p -Series.

↔ Stated in lecture as a fact, for use as a comparison series; cf. R 3.28.

Proposition 6.6.6

A nonnegative series converges when bounded

A series $\sum a_n$ of nonnegative terms ($a_n \geq 0$) converges if and only if its partial sums are bounded (R 3.24).

Proof 6.6.6

Proof by Monotone Convergence for a Series of Nonnegative Terms.

- (1) Since $a_{n+1} \geq 0$, the partial sums satisfy $s_{n+1} = s_n + a_{n+1} \geq s_n$, so $\{s_n\}$ is increasing.
- (2) If the partial sums are bounded, then $\{s_n\}$ is increasing and bounded, hence converges by the monotone convergence theorem (6.2.2); that is, $\sum a_n$ converges. Conversely, a convergent sequence is bounded. ■

[A5] cf. R 3.24; the partial sums increase; apply 6.2.2.

Proposition 6.6.7

The geometric series

For $|x| < 1$,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

This is the convergent half of Rudin's geometric-series theorem (R 3.26).

Proof 6.6.7

Proof by the Partial-Sum Formula of the Geometric Series.

- (1) For $x \neq 1$ the partial sums have the closed form

$$s_k = 1 + x + \cdots + x^k = \frac{1 - x^{k+1}}{1 - x}.$$

- (2) Since $|x| < 1$ gives $|x|^{k+1} \rightarrow 0$, we have $x^{k+1} \rightarrow 0$ and hence

$$s_k \xrightarrow{k \rightarrow \infty} \frac{1}{1 - x},$$

$$\text{so } \sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}.$$

■

[Direct] cf. R 3.26; uses $|x|^k \rightarrow 0$ for $|x| < 1$ (6.5.1).

Theorem 6.6.8**Comparison test for series**

- (a) If $|a_n| \leq c_n$ for all $n \geq N_0$ and $\sum c_n$ converges, then $\sum a_n$ converges.

- (b) If $a_n \geq d_n \geq 0$ for all n and $\sum d_n$ diverges, then $\sum a_n$ diverges.

This is Rudin's comparison test (R 3.25).⁵

Proof 6.6.8

Proof by the Cauchy Criterion of the Comparison Test.

- (1) For (a), let $\varepsilon > 0$. Since $\sum c_n$ converges, the Cauchy criterion (6.6.2) gives some $N \geq N_0$ with $\sum_{k=m}^n c_k < \varepsilon$ for all $n \geq m \geq N$. Then for such n, m ,

$$\left| \sum_{k=m}^n a_k \right| \leq \sum_{k=m}^n |a_k| \leq \sum_{k=m}^n c_k < \varepsilon,$$

so $\sum a_n$ meets the Cauchy criterion and therefore converges.

- (2) For (b), suppose instead $\sum a_n$ converged. Since $0 \leq d_n \leq a_n = |a_n|$, part (a) (with $c_n = a_n$) would force $\sum d_n$ to converge — contradicting that it diverges. Hence $\sum a_n$ diverges.

■

[A4] cf. R 3.25; bound the tail of $\sum a_n$ by that of $\sum c_n$.

The natural series to compare against is the geometric one. Comparing $\sum a_n$ to a geometric $\sum b_n$: if the ratio b_{n+1}/b_n is constant we are led to the *ratio test*, and if the root $\sqrt[n]{b_n}$ is constant,

⁵Convergence of a series is really a question of how fast the terms shrink, and comparison is the basic way to settle it: outrun a series known to converge and you converge too; be outrun by one known to diverge and you diverge. Most of the later tests are this one in disguise—they simply pick a convenient series, usually geometric, to measure against.

to the *root test*.

Theorem 6.6.9

Root test

Given $\sum a_n$, set $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Then

- if $\alpha < 1$, $\sum a_n$ converges;
- if $\alpha > 1$, $\sum a_n$ diverges;
- if $\alpha = 1$, the test is inconclusive.

This is Rudin's root test (R 3.33).

Proof 6.6.9

Proof by Comparison with a Geometric Series of the Root Test.

- (1) If $\alpha < 1$, choose β with $\alpha < \beta < 1$. Then $\sqrt[n]{|a_n|} < \beta$ for all large n , so $|a_n| < \beta^n$; as $\sum \beta^n$ converges, the comparison test (6.6.8) gives convergence.
- (2) If $\alpha > 1$, then $\sqrt[n]{|a_n|} > 1$ for infinitely many n , so $|a_n| > 1$ infinitely often and $a_n \not\rightarrow 0$; by 6.6.3 the series diverges. For $\alpha = 1$ the test decides nothing: both $\sum \frac{1}{n}$ (divergent) and $\sum \frac{1}{n^2}$ (convergent) have $\alpha = 1$. ■

[A1] cf. R 3.33; uses the comparison test (6.6.8).

Theorem 6.6.10

Ratio test

Suppose $a_n \neq 0$ for all large n . Then $\sum a_n$ converges if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, and diverges if there is N with $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ for all $n \geq N$. This is Rudin's ratio test (R 3.34).

Proof 6.6.10

Proof by Comparison with a Geometric Series of the Ratio Test.

- (1) If $\limsup |a_{n+1}/a_n| < 1$, choose β in between; then $|a_{n+1}/a_n| < \beta$ for $n \geq N$, so $|a_n| \leq |a_N| \beta^{n-N}$ for $n \geq N$. Comparison with the geometric series (6.6.8) gives convergence.
- (2) If $|a_{n+1}/a_n| \geq 1$ for all $n \geq N$, then $|a_n| \geq |a_N| > 0$ for $n \geq N$, so $a_n \not\rightarrow 0$ and the series diverges (6.6.3). ■

[A1] cf. R 3.34; uses the comparison test (6.6.8).

Definition 6.6.11

Power series and radius of convergence

A *power series* is a function defined by

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots .$$

Fix $x \neq 0$ and apply the root test (6.6.9)⁶ to $\sum c_n x^n$. With $\alpha := \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}$,

$$\beta := \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n x^n|} = |x| \alpha,$$

so the series converges when $\beta = |x| \alpha < 1$ and diverges when $|x| \alpha > 1$. Hence it converges for $|x| < R$, where

$$R := \frac{1}{\alpha} \quad (R := +\infty \text{ if } \alpha = 0, \quad R := 0 \text{ if } \alpha = +\infty).$$

R is the *radius of convergence*: the series converges on $(-R, R)$ and diverges for $|x| > R$; at $x = 0$ it is just $f(0) = c_0$, and the endpoints $|x| = R$ require separate analysis. This is Rudin's power-series setup and radius formula (R 3.38–R 3.39), restricted here to real x .

Example 6.6.12

The exponential series

For $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ the ratios have a limit, so they compute α (the valid case for the ratio shortcut):

$$\lim_{n \rightarrow \infty} \frac{1/(n+1)!}{1/n!} = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 = \alpha,$$

so $R = 1/\alpha = +\infty$: the series converges for every x . This function is $f(x) = e^x$. Compare Rudin's examples following the radius-of-convergence theorem (R 3.40).

⁶The root test is used here because it is strictly more general than the ratio test; for the precise relationship see the appendix, §6.A.1.

Appendix

Supplementary material—additional proofs, context, and other treatments; not part of the lecture proper.

Supplement 6.A.1

The root test subsumes the ratio test

For any sequence $\{c_n\}$ with $c_n \neq 0$,

$$\liminf_n \left| \frac{c_{n+1}}{c_n} \right| \leq \liminf_n \sqrt[n]{|c_n|} \leq \limsup_n \sqrt[n]{|c_n|} \leq \limsup_n \left| \frac{c_{n+1}}{c_n} \right|$$

(R 3.37). Hence whenever the ratio test settles convergence the root test settles it too, but not conversely; and if $\lim_n |c_{n+1}/c_n| = L$ exists then the root quantity tends to L as well, so the ratio test gives the same radius $R = 1/L$ (the converse fails — the radii can agree without the ratio limit existing). This is why 6.6.11 fixes R through the root quantity $\alpha = \limsup_n \sqrt[n]{|c_n|}$.

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